AE-SSS

Agentic Edge-AI Smart Surveillance System

> *"A camera can record a crisis. What if it could understand the context, know how serious it is, and help trigger the right response — without sending everyone's private data to the cloud?"*

PROPOSED ARCHITECTURE · IDEA / CONCEPT STAGE

Pettah Central Bus Stand, Colombo — conceptual deployment scenario

Cameras Watch. They Don’t Understand.

The real gap in busy public spaces isn’t a lack of cameras — it’s a lack of context.

TRADITIONAL CCTV

- ✗ Records events — doesn’t assess intent or severity
- ✗ Can’t coordinate an autonomous response
- ✗ Depends on a human noticing, watching multiple feeds
- ✗ Fatigue, cognitive overload, blind spots, delay

THE GAP THAT MATTERS

- ✓ Fast, context-aware understanding of what is happening
- ✓ A severity check — minor, moderate, or critical
- ✓ Rules that decide what response is safe
- ✓ A way to confirm whether the response actually worked

Conceptual scenario: imagine a crowded transit platform where an argument suddenly escalates, or a passenger collapses. Every second before someone understands what is happening is a second lost.

A Dense, Human, High-Stakes Environment

Project scenario: Pettah Central Bus Stand, Colombo — design-context assumptions, not measured statistics.

250,000+	**1,500+**	**85–95 dBA**	**3**
daily footfall (scenario assumption)	buses moving through daily	ambient noise — acoustic sensing must cope	languages in distress calls: Sinhala, Tamil, English

WHO THE PROPOSED SYSTEM AIMS TO PROTECT

🚶 **Commuters**

Daily riders, tourists & visitors

👩 **Vulnerable groups**

Women, children, the elderly

🚌 **Transit workers**

Drivers, conductors, vendors

🛡️ **Security teams**

Patrol officers, response coordinators

From Passive Surveillance to Governed Intelligence

AE-SSS — a proposed edge-first, agentic, closed-loop smart public-safety ecosystem

SENSE → UNDERSTAND → VALIDATE → ACT → VERIFY → LEARN

This is the physical-world loop. The intelligence layer that drives it is the Agentic AI OS — up next.

Edge-first

Critical decisions happen close to the event, not only in the cloud

Governed autonomy

AI recommends — it never acts without a safety check

Data minimization

Only meaningful metadata moves upstream, not raw streams

Closed-loop

The system checks whether its own action actually worked

Five Agents. One Governed Loop.

The Agentic AI OS: Monitor ⟶ Reason ⟶ Validate ⟶ Act ⟶ Learn

1

Monitor

Watches sensor data, flags anomalies & sensor health issues

2

Planner

Assesses severity — civic, moderate, or critical — and proposes a response

3

Validator

Checks policy, authorization & safety before anything can execute

4

Action

Executes only commands the Validator has approved

5

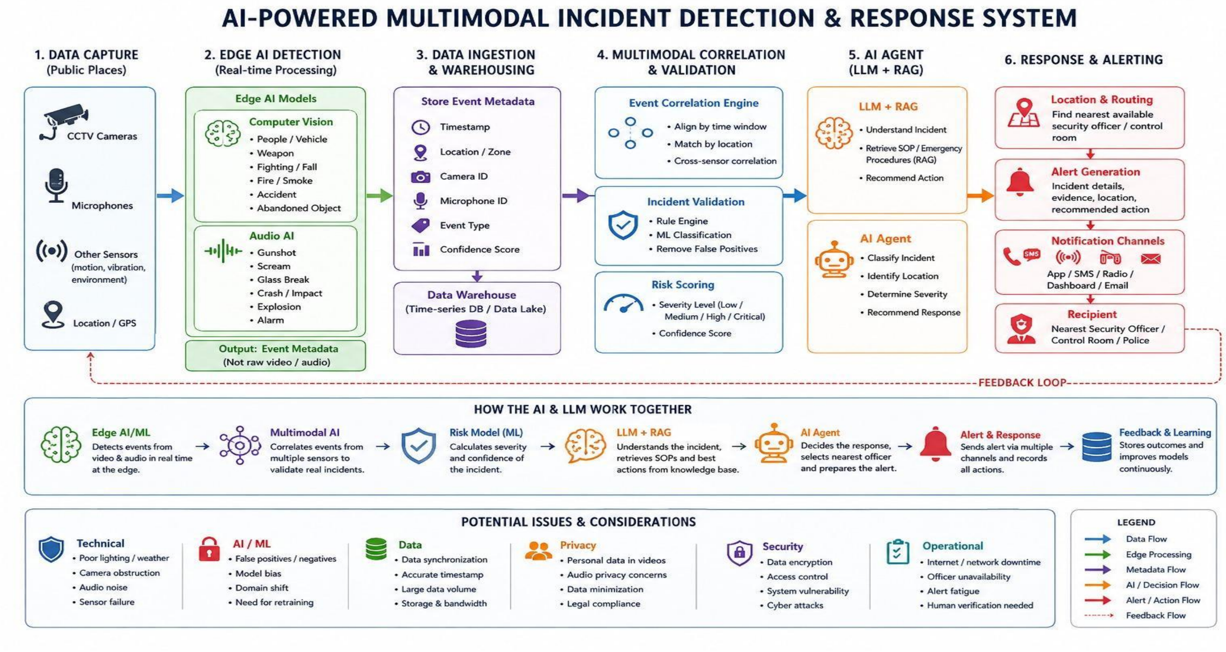
Learning

Observes the outcome and feeds it back into the loop

AI may recommend. The Validator decides whether that recommendation is allowed to execute. This is the core of ***governed autonomy*** — not uncontrolled AI.

A Five-Tier Sensing-to-Response Pipeline

Device → Edge → Network → Platform → Application — with a feedback path back to the physical world



DESIGN GOALS

- Send metadata, not raw video/audio, upstream
- Target: under 2 KB per event (design objective, not measured)
- Multi-modal: vision + acoustic + environmental telemetry
- Continue operating locally during network outages

Conceptual architecture diagram — illustrates the proposed design, not a deployed or measured system.

Not Every Event Deserves the Same Response

Concept walkthrough: a conceptual assault scenario moving through the governed loop

LEVEL 1 · CIVIC

Littering, spitting, minor obstruction

Active local deterrence (e.g. a spoken warning)

LEVEL 2 · MODERATE

Crowd surge, verbal altercation

Escalation / monitoring, based on context & policy

LEVEL 3 · CRITICAL

Assault, weapon threat, medical collapse

Silent tactical alert → emergency-response pathway

HOW ONE INCIDENT MOVES THROUGH THE LOOP

1. Sensors observe visual + acoustic cues
2. Monitor flags an unusual multi-modal event
3. Planner assesses severity as critical
4. Validator checks policy & authorization
5. Action sends a silent alert (not a loud alarm)
6. Feedback confirms whether it worked

Why silent, not loud? A loud warning could make an offender flee before help arrives.

Governed Autonomy: AI Recommends, Security Decides

Cybersecurity is 20% of the judging rubric — and it's built into the architecture, not bolted on

AI Recommendation → **Validator (policy + auth check)** → **Authorized Action**

Every recommendation must pass this gate before anything happens in the real world.

Identity & Access

- mTLS + X.509 device identity (proposed)
- Role-based access, least privilege
- Only approved commands can run

Human Oversight

- Human override for high-impact actions
- Structured logs — every action is traceable
- Safe stop when conditions are unsafe

Top RAID Risks

- Fake device → mitigated by mTLS & identity checks
- AI misclassification → mitigated by validation + override
- Connectivity loss → offline mode, store-and-forward

Why This Is Different — And How It Gets Built

COMPARED WITH A CONVENTIONAL PASSIVE-CCTV MODEL

Dimension	Traditional CCTV	AE-SSS (proposed)
Purpose	Passive recording	Context-aware intervention
Response	Manual, human-dependent	Governed, context-aware action
Privacy	Continuous recording	Data minimization + rolling buffer
Feedback	Usually after the fact	Closed-loop verification

FROM IDEA TO PILOT — A STAGED ROADMAP

1. Foundational sensing
2. Edge intelligence
3. Governed agentic loop
4. Security hardening
5. Controlled pilot
6. Institutional integration

Honest status: no working deployment yet. This is the engineering path — not a claim it already exists.

Smarter, Safer, More Sustainable Cities

SDG 11

Sustainable Cities & Communities

PRIMARY — a safer, more resilient public-transit environment

SDG 3

Good Health & Well-being

Faster awareness of medical emergencies & falls

SDG 8

Decent Work & Economic Growth

Safer working conditions for drivers, vendors, staff

SDG 13

Climate Action

Edge processing reduces unnecessary data movement

Sense the moment. Understand the risk. Act responsibly.

AE-SSS turns passive surveillance into a proposed privacy-aware, edge-intelligent, governed and closed-loop smart public-safety ecosystem.